

检索证明

经检索，以下 1 篇文献被《EI-Compendex》数据库收录的简要信息摘选如下：

Research on the Path of Folk Art: Utilizing Intelligent Mobile Devices for Art Education

Wang, Xiangqun (1)

Source:

International Journal of Information and Communication Technology Education, 20, 1, 2024

Database: Compendex

Document type: Journal article (JA)

Author affiliation: (1) Henan Finance University, China

Cited by in Scopus (0)

特此证明
(详细内容见附件)



扫描二维码查看报告



1. Research on the Path of Folk Art: Utilizing Intelligent Mobile Devices for Art Education

Accession number: 20243016759500

Authors: Wang, Xiangqun (1)

Author affiliation: (1) Henan Finance University, China

Corresponding author: Wang, Xiangqun(wxqun928@126.com)

Source title: International Journal of Information and Communication Technology Education

Abbreviated source title: Int. J. Inf. Commun. Technol. Educ.

Volume: 20

Issue: 1

Issue date: 2024

Publication Year: 2024

Language: English

ISSN: 1550-1876

Document type: Journal article (JA)

Publisher: IGI Global

Abstract: With the development of social economy, the globalization of the world has become an inevitable trend. However, the cultures of various countries and nationalities in the world are constantly communicating and merging with each other; the folk art in China has been greatly impacted, and the traditional culture with distinctive national characteristics has been rejected. Strengthening folk art education is needed not only to improve people's personality and cultivate people's aesthetic taste, but also to protect traditional culture and impart national spirit. With the rapid development of information technology relying on hand-held learning equipment, mobile internet technology, and network multimedia technology, life-long learning is possible. Using smart mobile devices to study the art teaching of folk art is conducive to in-depth exploration, applying the learning methods of folk art to our own study and life, and protecting and inheriting the infringed folk art. © 2024 IGI Global. All rights reserved.

Number of references: 23

Main heading: Engineering education

Controlled terms: ['Arts computing', 'Economic and social effects', 'Learning systems', 'Multimedia systems']

Uncontrolled terms: ['Art educations', 'Folk arts', 'Globalisation', 'Inevitable trends', 'Intelligent mobile device', 'Mobile Internet technology', 'Multimedia technologies', 'National characteristics', 'Social economy', 'Traditional cultures']

Classification code: ['723.2 Data Processing and Image Processing', '723.5 Computer Applications', '901.2 Education', '971 Social Sciences']

DOI: 10.4018/IJICTE.347668

Funding Text: This work was supported by the Research and Practice Project on Higher Education Teaching Reform in Henan Province in 2024: Research and Practice on the Cultivation Path of \u201CNew Liberal Arts\u201D Talents Based on Intangible Heritage Inheritance and Innovation

Compendex references: YES

Database: Compendex

Data Provider: Engineering Village

Compilation and indexing terms, © 2023 Elsevier Inc.

Research on the Path of Folk Art: Utilizing Intelligent Mobile Devices for Art Education

Xiangqun Wang
Henan Finance University, China

ABSTRACT

With the development of social economy, the globalization of the world has become an inevitable trend. However, the cultures of various countries and nationalities in the world are constantly communicating and merging with each other; the folk art in China has been greatly impacted, and the traditional culture with distinctive national characteristics has been rejected. Strengthening folk art education is needed not only to improve people's personality and cultivate people's aesthetic taste, but also to protect traditional culture and impart national spirit. With the rapid development of information technology relying on hand-held learning equipment, mobile internet technology, and network multimedia technology, life-long learning is possible. Using smart mobile devices to study the art teaching of folk art is conducive to in-depth exploration, applying the learning methods of folk art to our own study and life, and protecting and inheriting the infringed folk art.

KEYWORDS

Folk Art, Intelligent Mobile Device, Art Education

INTRODUCTION

Traditional folk art has been greatly impacted by the development of China's economy and the emergence of cultural diversity. Traditional culture with distinct ethnic characteristics is gradually being placed in a corner and faces the risk of being excluded. However, as a form of soft power, culture is increasingly valued by various countries in today's society. To stand firm on the global stage, a nation or country must possess its own unique cultural system and characteristics. Therefore, protecting folk art and culture is urgent. Therefore, protecting folk art and culture is urgent, and determining what methods should be used to protect them has become an issue of great concern. The best way to protect is through inheritance, and education is one of the most effective and sustainable ways to inherit culture. With the rapid development of handheld learning devices, mobile Internet technology and network multimedia technology, mobile learning and mobile education provide more possibilities for human lifelong learning (Zhang & de Bont, 2023). Against this background, this article applies the theory of microlearning based on mobile communication devices to the practical research on the reform of folk art educational models. We discuss the need to explore new forms of folk art learning, break through traditional teaching models, guide people to correctly use modern technological products, and engage in new learning fields, thereby accelerating the popularization of new learning models and the reform of folk art educational models. This article helps to encourage educational institutions to explore new models of folk art education and promote their popularization.

DOI: 10.4018/IJICTE.347668

This article published as an Open Access article distributed under the terms of the Creative Commons Attribution License (<http://creativecommons.org/licenses/by/4.0/>) which permits unrestricted use, distribution, and production in any medium, provided the author of the original work and original publication source are properly credited.

The innovation of this article lies in the comprehensive collection of and research on woodblock New Year paintings from multiple regions, showcasing the representativeness of folk art styles and schools from different regions. By comparing and analyzing the characteristics of these New Year paintings, the diversity and regional characteristics of traditional Chinese folk New Year paintings have been revealed. Meanwhile, by digitizing and analyzing these New Year paintings, new ideas and methods can be provided for the protection and inheritance of cultural heritage.

LITERATURE REVIEW

In the long process of human history in our country, the ethnic groups in each region have formed their own unique local cultural genes in the long-term struggle against the powerful nature for their own survival and development. The natural environment and social ecological environment of each nation are different, and folk art is an important part of an important part of that difference (Wei, 2021). These are the embodiment of the creativity and collective wisdom of the working people of a nation, and represent a visual art that integrates the ideology, morality, consciousness, and aesthetic taste in order to satisfy the working people's own life and spiritual pursuit (North et al., 2024).

There are countless works of folk art. The earliest works can be traced back to the period of "Three Sovereigns and Five Emperors." The earliest works can be traced back to the period of "Three Sovereigns and Five Emperors" and include grottos from the period and portrait bricks of the Han Dynasty, among others (Kitto et al., 2024). Another example is the silk paintings unearthed in Mawangdui, Changsha. These works of art with distinctive ethnic characteristics are all evidence of the origin and development of China's folk art culture. The cultural diversity contained in it has the following two aspects. First, compared with the orthodox art form of the ancient court, folk art is made by working people without formal training. Second, in terms of contemporary art, Minwen art is opposed to contemporary professional art. But culture has its value, and so does folk art, which best reflects the values and beliefs of the people in that period (Rahmatjonzoda, 2024). At present, many people hold a negative view of folk and think it means outdated or old-fashioned (Wang, 2024). In fact, people who have this idea do not understand the positive value of folk art. And Because folk art arises from nature and life, many people can easily connect to it and it has popular appeal.. It is because it is taken from nature and life, so it is easy to be recognized and loved by the public (Kalota, 2024). Due to the simple form of national art and the distinctive national characteristics in the content, it has a good educational effect (Wu & Sensai, 2021). National art forms generally appear at the same time as national activities, so they are more easily accepted by people and gradually become an important carrier of national culture, which promotes the formation of national identity, such as the common clay sculpture art, shadow play, and paper-cut art, among others (Wang et al., 2021).

Alkaf et al. (2024) explored the relationship between the inheritance of traditional culture and the development of art education, proposing measures to promote traditional culture in art education. Gong (2024) discussed oral folk art as part of the aesthetic education of schoolchildren in the North Caucasus, emphasizing the formation of specific aesthetic and artistic patterns in individuals. Ochilova (2024) highlighted the value of Chinese traditional culture in art and design education, emphasizing the importance of embedding traditional culture in education. Shodieva (2024) discussed the digital inheritance of folk art based on cognitive psychology, using the Guangfu Furniture Museum as a case study. Gurdjieff's teachings at the Rochester Folk Art Guild engage intelligence through writings, movements, and music. The application of artificial intelligence technology in art teaching is also explored, comparing different algorithms for teaching methods. While some argue that traditional art education is failing and may not be necessary for aspiring artists, others question whether artificial intelligence will replace traditional artists in the future. The impact of artificial intelligence on art is also discussed, with advancements in technologies such as neural networks and machine learning (Sade et al., 2024). Overall, the literature suggests a complex relationship between folk art, intelligence, and education, with discussions on the importance of traditional culture in art education, the role of

digital technologies in preserving folk art, and the potential impact of artificial intelligence on the art world (Li, 2024).

MATERIALS AND METHODS

Folk Art and Its Aesthetic Characteristics

Folk art is created by the people, starting from the needs of life and making things for their own use. As an art form, folk art is created and developed by the people. It is the response of the people's needs and pursuits. It is deeply loved by the people (Liu, 2020). In the process of historical development, folk art resources are the crystallization of human wisdom. Folk art contains the traditional culture of various ethnic groups and is the source and foundation of art teaching. The content of folk art resources is rich and diverse, including practical art resources, and art resources derived from folk activities and for the purpose of enjoyment. Specifically, it includes: paper-cuts, New Year pictures, prints, shadow puppets, batik, embroidery, ceramics, buns, tiger-head shoes, and more (Gryglewski et al., 2020).

Humans have created culture, and the formation of all cultural phenomena is made up of people and their way of life and behavior. The so-called “one side’s soil and water nurtures the other side’s person” means this. Folk art represents people's ways of thinking and behavioral norms, including scientific concepts, moral concepts, values, political concepts, aesthetic awareness, folk psychology, lifestyle, religion, and other factors (Liu, 2023). The culture of a nation is deeply hidden in the fertile soil of national life, but it can present its own characteristics in various aspects. Folk art is the art of laborers. It is rooted in the countryside and inherited from generation to generation, showing a clear context for the development of a national culture. Folk art acts on people's spiritual life and material life in the form of popular aesthetics, and it educates people on morality, history, customs, production, and cultural knowledge with rich content and has become the most popular textbook.

Folk art is an art system with the longest history, the most authors, the widest circulation, and the greatest value in the national art of our country. Its shape emphasizes subjective imagination, free expression, and emotional expression. Its function is both practical and educational, and it is a product of spiritual materialization. Folk art is rooted in the folk, and the folk are rarely influenced by external political situations. The simple and honest folk customs nourish folk art. The open-minded, optimistic, honest, and diligent characters of laborers are reflected in the art (Rochovská & Švábová, 2021).

The Educational Value of Folk Art

Cultural knowledge is the source of folk art and its unique characteristics. Folk art contains the cultural connotation of the nation and the sum of knowledge in other fields. It is not only the cultural knowledge that contemporary people need to have but also an entry point. People in society should learn to constantly enrich their knowledge. The connotation of folk art affects all aspects of our life and study and also has a profound impact on our spiritual world. Folk art is an indispensable part of art education. It permeates our working life and is scattered around us. It is of great help to the understanding and appreciation of the traditional culture of our nation. Folk art contains the hardships of the working people. The good wishes for life and the education of people's ideology and morality also play an irreplaceable role in the content of other disciplines. Folk art not only has the function of improving people's aesthetic ability but also has different effects on the development of people's creative thinking and intelligence, as well as people's hands-on ability. Folk art education is proposed under the requirements of China's new curriculum standards. It combines the cultivation of people's art skills with the improvement of humanistic quality. Folk art education focuses on cultivating people's aesthetic emotions and creativity. This shows that China's folk art represents one of the best ways for schools to implement quality education.

With the rapid development of technology, smart mobile devices have become an indispensable part of our lives. In this digital era, how to combine folk art with intelligent mobile devices for effective art education has become a problem that we need to explore. Folk art, as an important component of traditional Chinese culture, carries rich historical and cultural information. However, with the rapid development of society, the preservation of folk art faces many challenges. The popularization of smart mobile devices has brought new opportunities for the dissemination and education of folk art. Social media platforms such as WeChat and Weibo can be leveraged to widely spread the charm of folk art. Through the creation of official accounts, online exhibitions, online competitions, and other forms, more young people are attracted to and participate in the preservation of folk art. Online education platforms can be used to launch a series of folk art courses to launch a series of folk art courses, inviting artists and scholars for online teaching, providing professional guidance and communication, and promoting the popularization and improvement of folk art. In order to maintain the core elements of folk art, modern aesthetics and technological means can be combined to innovate educational content, making it more in line with the tastes and needs of contemporary users. In the development and dissemination of educational resources, it is important to attach importance to intellectual property protection and establish a sound intellectual property protection mechanism to prevent infringement (Zhu, 2023).

Microlearning of Folk Art Education Under Intelligent Mobile Technology

With the popularization of mobile Internet technology and the rapid development of mobile device terminals, technological changes have profoundly affected all aspects of society. Microlearning based on mobile communication technology has provided new ideas for folk art education. The application of technology in teaching has gradually become a new normal, which has expanded and extended the learning space and enriched the learning resources. It has the characteristics of smallness, simplicity, and speed. With knowledge content, it caters to the needs of learners to acquire knowledge with a relaxed and pleasant attitude in a short period of time, and refined and short learning content can be obtained, circulated, and disseminated on portable electronic devices anytime, anywhere (Wei, 2021). Microlearning is a kind of informal learning. Learners can easily acquire, store, generate, and circulate knowledge through mobile communication devices and complete the learning experience in a relaxed and happy state of mind. Bringing the concept of microlearning into folk art education and carrying out popular activities on art knowledge not only caters to people's interests but also conforms to the needs of the times.

Constructivist learning theory is an indispensable theoretical basis for the study of micro-learning. It emphasizes fully respecting the individuality of learners and encouraging students to actively participate in the learning process. Constructivism advocates that the teaching process should be people-oriented, fully reflect the subject status of students, and pay attention to the learner's personality characteristics and interest orientation. Figure 1 shows the development of learning methods in the Internet era, from face-to-face teaching to micro-lectures, MOOCs, and flipped classrooms, to the emergence of maker-style education and mobile learning. The construction of a teaching platform based on intelligent mobile devices supported by mobile technology is conducive to the integration of market production, university teaching and research, scientific research institutions and other resources, and it is also conducive to promoting the integration and communication between teaching, enterprise production, and scientific research, as well as to optimization.

Content refers to the mobile learning resources provided to learners in a mobile learning course. *Computation* means that mobile learning obtains data from learners, processes that data, and gives corresponding feedback to learners. *Acquisition* means that learners share their ideas and insights from the outside world with other learners through the design of mobile learning resources in the form of audio, video, and text. *Communication* refers to the interaction of learners with tutors or peers. In microlearning, learners can select relevant microcontent to learn new knowledge according to their own interests and previous knowledge and experience and, at the same time, establish collaborative

Figure 1. Learning Methods in the Internet Age

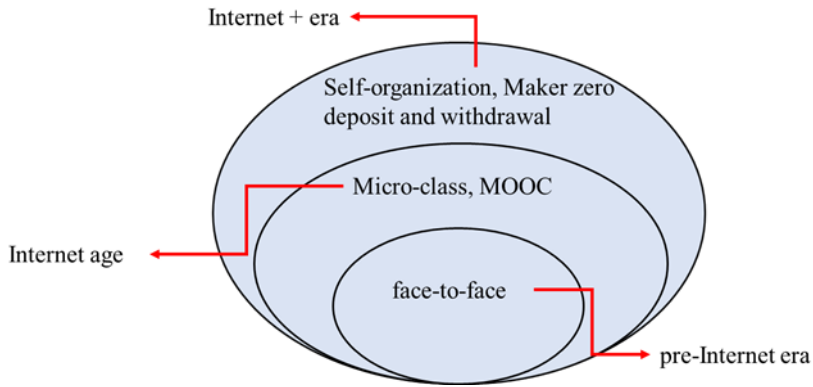
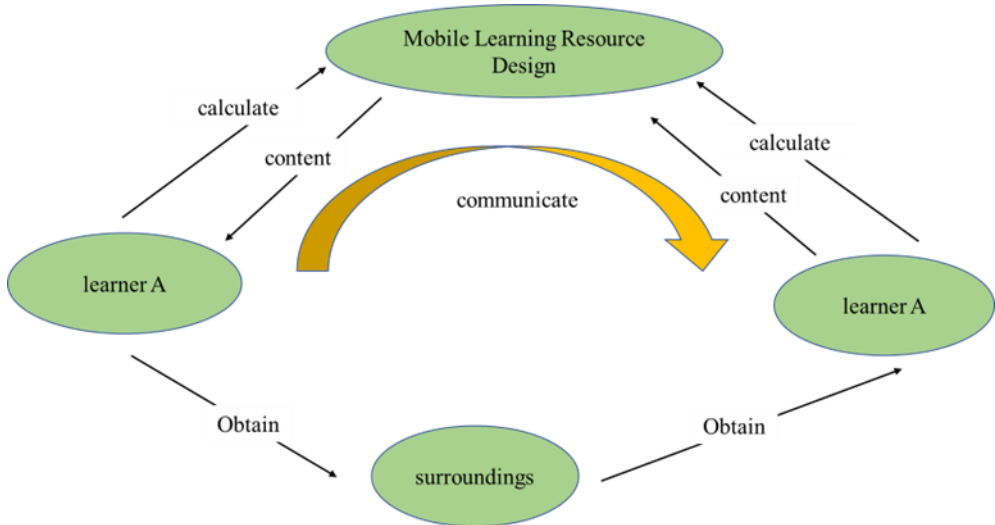


Figure 2. Mobile Learning 4C Model



relationships with other learners through the Internet to complete their meaning construction. Figure 2 presents the 4C model related to constructivism, according to the literature, consists of contextualizing, connecting, constructing, and cognitivizing.

Classification of Folk Art Based on Dual-Channel Neural Network

The common types of folk art include painting, embroidery, and paper-cutting. This section mainly takes painting in folk art as an example for classification research. The definition of the art painting image classification method is: a method for classifying the attributes of each painting image, and the attributes include which painting style and genre the art painting image belongs to. The classification method should have the following characteristics: mainly focus on the overall content of the image rather than concentrating on specific details, that is, classify the attributes of the entire image. Taking New Year pictures as an example, it is mainly used in the Spring Festival, symbolizing auspiciousness and blessing. As an important part of China's intangible cultural heritage, there are many types of New Year pictures due to different regions, and their content is mostly based on secular social life

and covers everything. Chinese folk woodblock New Year pictures are widely distributed in regions. After the mid-Ming Dynasty, workshops for carving New Year pictures spread almost all over the country. According to the collected works, in addition to Heilongjiang, Jilin, Xinjiang, Ningxia, Tibet, Inner Mongolia, and other places, there used to be handicraft workshops everywhere. New Year pictures have gradually formed different artistic styles and unique regional characteristics in the long history, such as Tianjin Yangliuqing, Henan Kaifeng Zhuxian Town, Jiangsu Suzhou Taohuawu, Shandong Weifang Yangjiabu, Sichuan Mianzhu, Hebei Wuqiang, and Guangdong Province Foshan. The Tantou woodblock New Year pictures in Fengxiang, Zhangzhou, Fujian and Shaoyang, Hunan are all well-known and have their own characteristics.

With the continuous development of deep learning technology, using large-scale datasets for pre-training has become an important means to improve model performance. Among them, the ImageNet dataset has become the cornerstone of many computer vision tasks due to its rich categories and large scale. For the classification problem of art painting images, using the ImageNet dataset for pre-training is expected to significantly improve the classification performance of the model. The ImageNet dataset contains a wide range of image sources, including various changes in perspective, lighting, background, and object pose, which helps to improve the model's generalization ability. It uses a pre-trained model on ImageNet as the starting point and then fine tunes the last or fully connected layer to adapt to the classification task of art painting images. This can optimize for specific tasks while maintaining the existing knowledge of the model. Due to the rich style and color of art painting images, data augmentation techniques such as style transfer and color adjustment can be used to generate more diverse training samples, further improving the model's generalization ability. A pre-trained model on ImageNet can be used as a feature extractor to extract features from art painting images. These features are more likely to capture visual patterns similar to the ImageNet dataset, which helps improve classification accuracy. In some cases, there may be art painting image datasets with significant differences in distribution compared to the ImageNet dataset. At this point, the pre-trained model can be used as a feature extractor, combined with cross-domain learning algorithms, to enable the model to better adapt to these differences. We can consider combining pre-trained models with other methods related to art painting image classification, such as style transfer and feature matching, to form a hybrid model in order to better preserve the unique style and characteristics of art painting while maintaining classification accuracy.

Figure 3 shows the two-channel two-path network architecture for art painting image classification. Convolutional neural network (CNN) is a deep learning algorithm. Figure 4 shows a neural network structure diagram. Like the traditional neural network, it is also a hierarchical network. The three parts of convolution, pooling, and activation are the main structures of CNN.

The data inputs a part of the information into the next layer through a layer of convolution and then conducts a more in-depth analysis of this part of the information; the matrix will be further deepened each time the convolution is performed. A convolution process can be expressed as:

$$conv_{x,y} = \sum_i^{n-m} w_i v_i \quad (1)$$

where w is the kernel weight, v is the brightness value of the image, and $n-m$ is the size of the convolution kernel. The output of the convolutional layer is shown in the formula:

$$x_j^{l-1} = f(u_j^l) \quad (2)$$

where u can be expressed as:

$$u_j^l = \sum_{i \in M_j} x_j^{l-1} * W_{ij}^l + b_j^l \quad (3)$$

Figure 3. Two-Channel Two-Path Network Architecture for Art Painting Image Classification

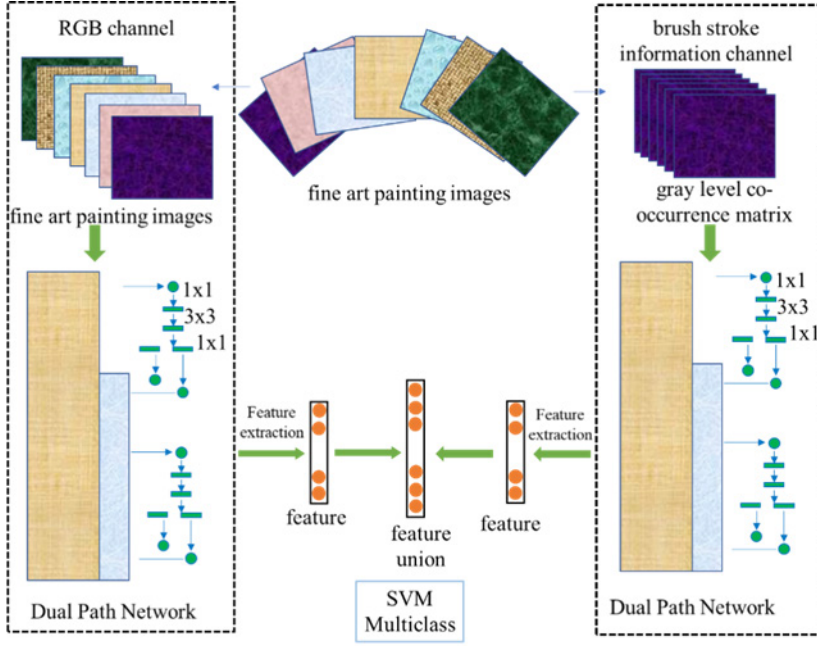
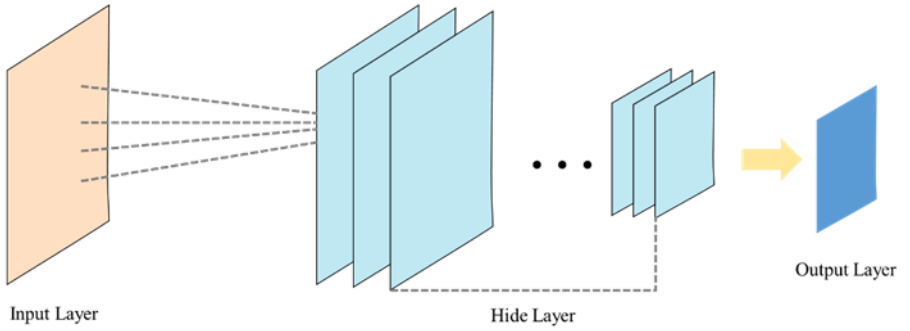


Figure 4. Neural Network Structure



After the pooling layer, the number of output values of the convolutional layer can be simplified, thereby reducing the complexity of the entire network in order to avoid the problem of overfitting. The expression of the pooling layer is as follows:

$$A_K^l(i,j) = \left[\sum_{x=1}^f \sum_{y=1}^f A_k^l(s_0 i + x, s_0 j + y)^p \right]^{\frac{1}{p}} \quad (4)$$

The value of p determines the meaning of the pooling layer. When p is 1, it is mean pooling, and when p tends to infinity, it is maximum pooling. Currently, the latter is the most used, that is, maximum pooling. After the convolution, the nonlinear operation is performed through the activation function. The operation process is shown in the formula:

$$Z_{x,y} = h(\sum_i^{n-m} w_i v_i + b) \quad (5)$$

Image processing uses the convolution method. The entire operation is linear, but the sample is not linear; if it is a linear function, the nonlinear problem cannot be solved, so an activation function needs to be added here. Three commonly used activation functions are listed below, formula (6) is the sigmoid function, formula (7) is the tanh function, and formula (8) is the relu function:

$$s(z) = \frac{1}{1 + e^{-z}} \quad (6)$$

$$t(z) = \tanh(z) \quad (7)$$

$$r(z) = \max(0, z) \quad (8)$$

Some optimizations of the neural network are also required: there are four improvement methods from a model perspective: weight decay, dropout, batch regularization, and LRN.

The weight decay method can effectively deal with the occurrence of overfitting by adding regularization to the original data. The regularization term is shown in formula (9):

$$C = C_0 + \frac{\gamma}{2n} \sum w^2 \quad (9)$$

Among them, C_0 is the original data, n is the sample size, and the λ coefficient is to balance the weight. Squaring and summing all parameters w , dividing by n is a complete L2 regularization process. The reference to dropout is to avoid overfitting during training.

1. Calculate the mean:

$$u\beta \leftarrow \frac{1}{m} \sum_{i=1}^m x_i \quad (10)$$

2. Calculate the variance of the data:

$$\sigma_\beta^2 \leftarrow \frac{1}{m} \sum_{i=1}^m (x_i - u\beta)^2 \quad (11)$$

3. Normalization:

$$\hat{x}_i \leftarrow \frac{x_i - u\beta}{\sqrt{\sigma_\beta^2 + \varepsilon}} \quad (12)$$

4. Scale shift:

$$y_i \leftarrow \gamma \hat{x}_i + \beta = BN_{\gamma, \beta} \quad (13)$$

Local response normalization is a process that changes the values of neurons in the network, increasing the training speed of the entire model. The LRN expression is shown in the formula:

$$b_{x,y}^i = \frac{a_{x,y}^i}{(k + \alpha \sum_{j=\max(0,i-n/2)}^{\min(N-1,i+n/2)} (a_{x,y}^j)^2)^\beta} \quad (14)$$

This is mainly due to the effective combination of image feature reuse and exploration of new features in the dual path network, which greatly improves classification performance.

In this paper, 2048-dimensional vectors (features) are extracted from each channel, and the 4096-dimensional features formed by concatenation constitute the input of the SVM classifier. Specifically, 2048 is the number of feature maps extracted from the last fully connected layer.

Among them, the residual path is mainly composed of residual network. The residual learning in the residual network can not only improve the performance of the model under the deepening of the network depth but also ensure the stability of network training. The residual network transfers part of the original input feature map directly to the next layer without going through the convolution layer and nonlinear activation layer. This is like establishing many interconnected information highways in the deep network, reducing unnecessary information loss in the middle. Specifically, the residual network changes the learning objective, that is, instead of learning the final output, it learns the residuals. Therefore, the residual network effectively avoids the loss caused by the convolutional layer during information transfer. The residual network protects the integrity and efficient transmission of information to a certain extent through the skip structure. Dense connection paths are mainly composed of dense convolutional materials. For each convolutional layer in a dense convolutional network, the feature maps of all convolutional layers can be used as input, and its own feature maps can be used as input to all subsequent layers. The concept of shared features and arbitrary inter-layer interconnection proposed by dense convolutional networks effectively avoids the problem of difficult optimization caused by gradient disappearance/explosion in deep networks during training. At the same time, due to the optimization of the dense convolutional network structure and parameter sharing, the size and computational complexity of the deep model are also reduced. A big advantage of dense convolutional networks is that they improve information flow and gradients throughout the network, which makes deeper neural networks easier to train. It is worth noting that the dense connection has a regularization effect, which can effectively avoid the overfitting problem caused by the small number of samples in the training set.

It is very reasonable to use a dual path network (DPN) for classification. By introducing two independent propagation paths in each module, DPN can increase the depth of the network, thereby improving its representational and learning abilities. This helps the network better capture complex features and structures in images, improving classification performance. The design of DPN allows different paths to process features of different scales, and then fuse these features. This multi-scale feature fusion capability enables the network to simultaneously focus on local details and global context, which helps improve classification accuracy. Compared to simply increasing network depth, DPN designs a more effective network structure that minimizes the number of parameters while increasing network depth. This can reduce the computational and storage overhead of the model while maintaining high performance, making DPN more feasible in practical applications. By introducing multiple paths and multi-scale feature fusion, DPN can better handle issues such as noise, deformation, and occlusion in input images; improve the robustness and generalization ability of the network; and perform more robustly in various complex scenes. Based on the above reasons, DPN, as a network model for classification tasks, can effectively improve classification performance through its deeper network structure, multi-scale feature fusion, parameter efficiency, robustness, and generalization ability.

The advantages of DPN compared to other architectures are mainly reflected in the following aspects. DPN adopts a dual path design, which allows the network to learn image features from different perspectives simultaneously. In some tasks, this multipath design can better capture features at different levels and scales in the image, thereby improving the performance of the model. The design of DPN allows different paths to process features of different scales and fuse them together. This multi-scale feature fusion capability can better handle local details and global information in images, thereby improving the model's understanding of images. Compared to some other deep network architectures, DPN minimizes the number of parameters while increasing network depth. This enables DPN to maintain high performance while reducing the computational and storage overhead of the model, making it more suitable for resource limited environments or scenarios that require deployment on mobile devices. Due to the multipath design of DPN being able to better handle issues such as noise, deformation, and occlusion in input images, DPN has strong robustness and generalization ability. In practical tasks, this ability can help the model better adapt to various complex scenarios, improving the accuracy and stability of the task. In summary, due to the advantages of DPN in multipath design, multi-scale feature fusion, parameter efficiency, robustness, and generalization ability, it is more suitable compared to other architectures and can provide better performance.

Collaborative Filtering Recommendation Algorithm for Folk Art

Recommender systems, also known as information filtering systems, are used to predict users' ratings or preferences for items. The recommendation system uses different recommendation algorithms to analyze data such as recommended objects, user behavior, and user information; forms recommendation rules; calculates and obtains recommendation sets; and makes personalized recommendations for target users. Collaborative filtering recommendation algorithm is a mainstream and widely used recommendation algorithm. To predict users' preferences for items and complete personalized recommendations. Because the algorithm has the advantages of simple algorithm and flexible use, it has been well received by the majority of users. The following two algorithms are briefly introduced.

- The user-based collaborative filtering algorithm mainly considers the similarity between system users. The algorithm finds user behavior rules through user behavior logs and converts the obtained data into computer-identifiable data. Then, based on the obtained user preference data, a similarity algorithm is employed to determine “neighbor groups.” Finally, the target users are recommended through the neighbors' preferences.
- The item-based collaborative filtering algorithm calculates and obtains the item set with the highest similarity with the target item through historical user evaluation data, predicts the target user's rating of the item, and recommends the N items with the highest rating for the user.

There are many ways to calculate similarity between users. Among them, the cosine algorithm is more commonly used, and its formula can be expressed as:

$$w_{uv} = \frac{|N(u) \cap N(v)|}{\sqrt{|N(u)||N(v)|}} \quad (15)$$

After the neighbors of the target user are obtained, we can predict the user's recommendation score for the item that has not been browsed. The recommendation formula is as follows:

$$P_{u,j} = \bar{R}_u + \frac{\sum_{v \in N} w_{uv} \times (R_{v,j} - \bar{R}_v)}{\sum_{v \in N} |w_{uv}|} \quad (16)$$

In the formula, P is used to represent the predicted rating value of user u for item j , R is the average score of user v to the evaluated recommended objects, and N is the set of neighbors of target user u . By sorting the predicted score value P , the K items with the highest score value are obtained to make a recommendation for the user.

RESULTS AND ANALYSIS

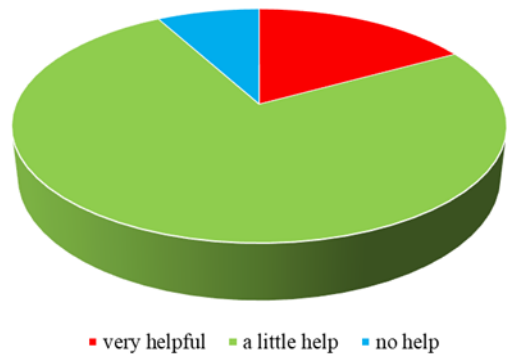
Experimental Design and Testing Methods

Since there is no existing database related to folk art at present, in the experiment of this paper, we first collect the materials of New Year pictures from different regions to construct a simple data set and then divide the data set into training set and test set to conduct folk art research. We collected Tantou woodblock New Year pictures from Yangliuqing in Tianjin, Zhuxian Town in Kaifeng in Henan, Taohuawu in Suzhou in Jiangsu, Yangjiabu in Weifang in Shandong, Mianzhu in Sichuan, Wuqiang in Hebei, Fengxiang in Foshan in Guangdong, Zhangzhou in Fujian, and Shaoyang in Hunan. The collection totaled 1000 samples of New Year paintings. These samples cover different eras and styles from the Ming and Qing dynasties to the contemporary era. The differences in fine art painting images across the different categories are very subtle. Although all of these fine art painting images belong to the theme of New Year pictures, they belong to a completely different style category in the style classification of folk art painting images. Tianjin Yangliuqing is known for its delicate lines and bright colors, often depicting folk life scenes and traditional cultural elements. Zhuxian Town in Kaifeng, Henan Province pays attention to the depth and rendering techniques of the brush and ink, and the works often have a folk style that is full of vitality. Suzhou Taohuawu in Jiangsu Province is characterized by exquisite depiction and smooth lines, with works mostly featuring traditional themes of mountains, rivers, flowers, birds, and characters. Yangjiabu in Weifang, Shandong is mainly characterized by a simple and lively style and bright colors, with works mostly featuring myths, legends, and folk stories. Sichuan Mianzhu is known for its unique folk style, often displaying strong local characteristics and cultural atmosphere. Hebei Wuqiang is known for its unique lines and decorative features, and his works often have strong decorative and expressive power. Fengxiang in Foshan, Guangdong emphasizes the inheritance and development of traditional themes, and its works are often creative and decorative. Zhangzhou, Fujian is known for its unique artistic style and diverse subject matter, often depicting local folk customs and natural landscapes. Shaoyang, Hunan Province, is themed around its unique regional characteristics and ethnic culture, with many works featuring blessings and folk legends. The New Year paintings in these regions inherit their unique folk art styles and cultural traditions, providing rich sample resources and cultural backgrounds for research.

This paper adopts common data augmentation methods (such as cropping, mirroring, and rotation, etc.) to increase the diversity of samples during training (Abayomi et al., 2021). Specifically, we randomly crop each sample of New Year paintings to a size of 224x224 pixels and flip them horizontally and vertically with a probability of 0.5, while performing a random rotation of ± 15 degrees. These operations can increase the diversity of training samples and help the model learn more robust features. We use a DPN as the underlying network architecture and initialize it with pre-trained weights. During the training process, we used a stochastic gradient descent (SGD) optimizer with an initial learning rate of 0.01 and adopted a learning rate decay strategy to reduce the learning rate to half of the original at the end of each epoch. For batch size, we set it to 32 when training on a single GPU, and for the deepest neural network with 131 layers of DPN, the batch size for a single GPU is set to 24.

For complex datasets and model architectures (such as 131 layer DPN), addressing overfitting is crucial. Here are some common methods to alleviate overfitting. By performing random transformations, rotations, cropping, scaling, and other operations on training data, more diverse data samples can be generated, which helps the model learn more generalized features. During the training process, randomly set the output of a portion of neurons to 0 to reduce the dependency

Figure 5. The Impact of Mobile Learning Resource Design on Folk Art Learning



relationship between neurons, reduce the model's dependence on certain specific features, and thus alleviate overfitting. By monitoring the performance of the validation set, stop training before the model performance begins to decline to prevent overfitting. Combine models obtained from multiple training processes, such as voting or averaging, to obtain more robust and better generalization performance results. Standardizing the input of each small batch of data helps to accelerate model training and can to some extent reduce overfitting. Prune or compress complex models to reduce model complexity and thus reduce the risk of overfitting. These methods are usually used in combination to select appropriate combinations based on specific tasks and model characteristics in order to reduce overfitting and improve model generalization ability.

Comparison and Analysis of Experimental Results

In addition to designing the functions and interfaces of the folk art learning resource design based on the Clark Quinn 4C model, this paper also conducts teaching applications. The teaching application of folk art mobile learning resource design based on Clark Quinn's 4C model is aimed at students in a school's art class. The teaching application results of middle school physics mobile learning resource design based on Clark Quinn's 4C model are obtained through interviews. Figures 5 and 6 show the impact of mobile learning resource construction on folk art learning and learning interest, respectively.

It can be seen from the survey data that most learners believe that the design of mobile learning resources for folk art based on the Clark Quinn 4C model is helpful for folk art learning and helps to better understand folk art related knowledge (Li & Zhang, 2020). Learners generally believe that the design of mobile learning resources for folk art based on the Clark Quinn 4C model has a certain interesting performance to stimulate learners' interest in learning folk art. It can be seen that mobile learning will change the learning style of learners to a certain extent and encourage them to use mobile technology to learn the knowledge of various disciplines. Figure 7 shows the classification performance comparison results of 50 layers of ResNet stroke representation.

Figure 8 shows the classification performance comparison results of 131 layers of DPN stroke representation methods. Through the above experiments, it can be clearly found that GLCM can extract the information of strokes better than Gram matrix. However, this paper cannot fully conclude that GLCM can also exert better classification performance in non-deep learning methods. Therefore, in the following experiments of this paper, we compare the classification performance achieved on other existing statistical methods or machine learning methods with the Gram matrix and using GLCM as input (Zhong et al., 2020).

GLCM can reveal texture features in images. When using GLCM in four directions, pixel relationships in the horizontal and vertical directions, as well as in the 45 ° and 135 ° directions, can be

Figure 6. The Influence of Mobile Learning Resource Design on Interest in Learning Folk Art

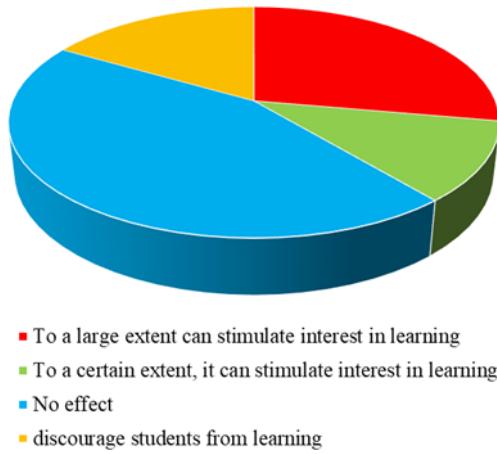
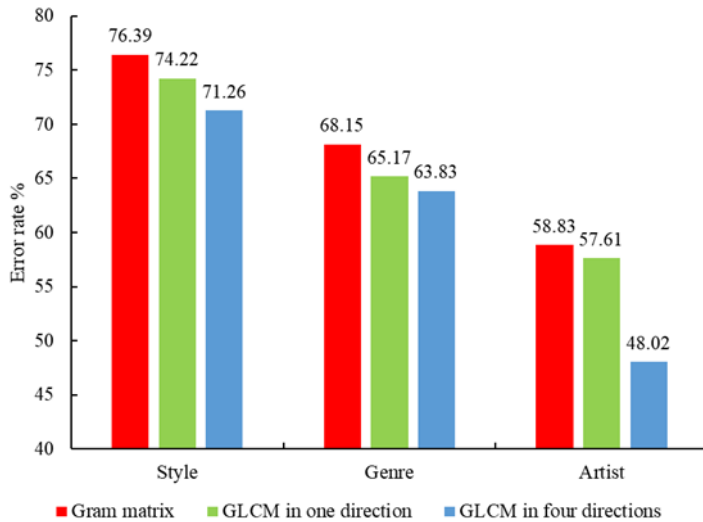


Figure 7. Comparison of Classification Performance With 50-Layer ResNet Stroke Representation Method



considered separately. These four directions of GLCM can capture texture information from different directions in the image, thus more comprehensively describing the characteristics of strokes. The four directional GLCM can capture the changes of strokes in different directions, which is particularly important for analyzing paintings with multiple directional changes. By comprehensively considering the texture information from four directions, the model can better understand the directional features of strokes. GLCM in different directions can highlight texture details in different directions. This is very beneficial for paintings with significant texture changes in specific directions, such as certain impressionist or abstract art works. The four directions of GLCM complement each other, providing a more comprehensive texture description. By combining this information, the model can more accurately analyze the complexity and diversity of strokes, thereby improving the accuracy of classification. Merge GLCM features from four directions together to form a unified feature vector. Simple averaging or weighted averaging methods can be used, or more complex fusion strategies

Figure 8. Comparison of Classification Performance with 131-Layer DPN Stroke Representation Method

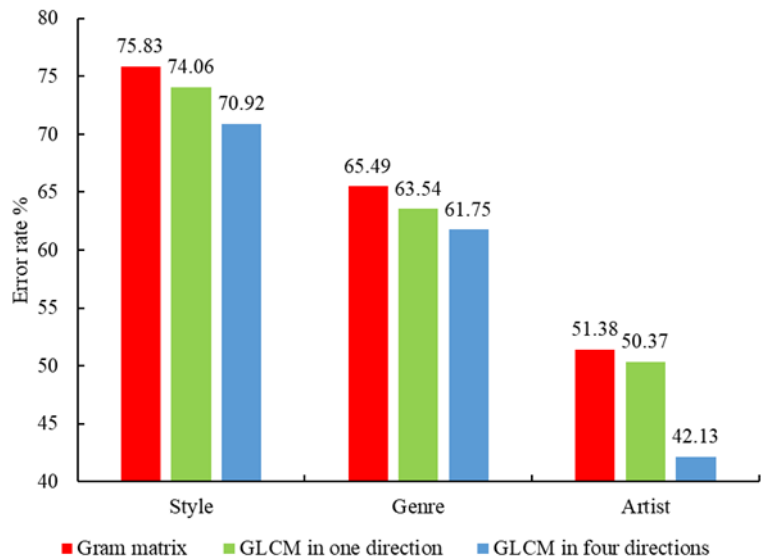
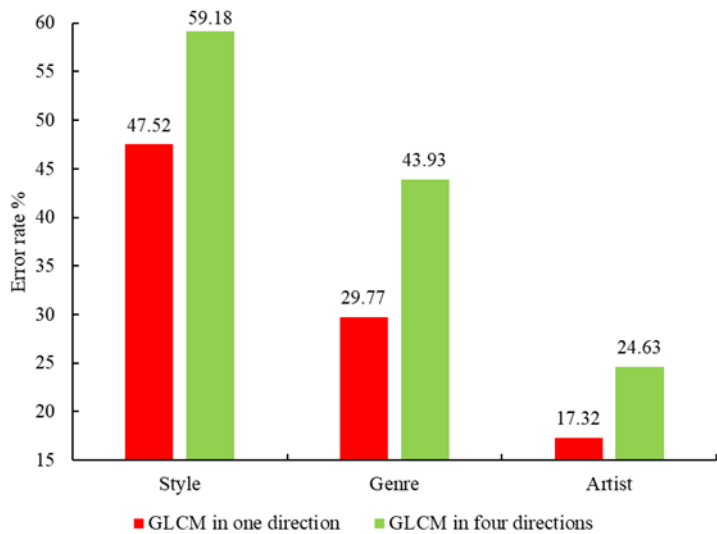


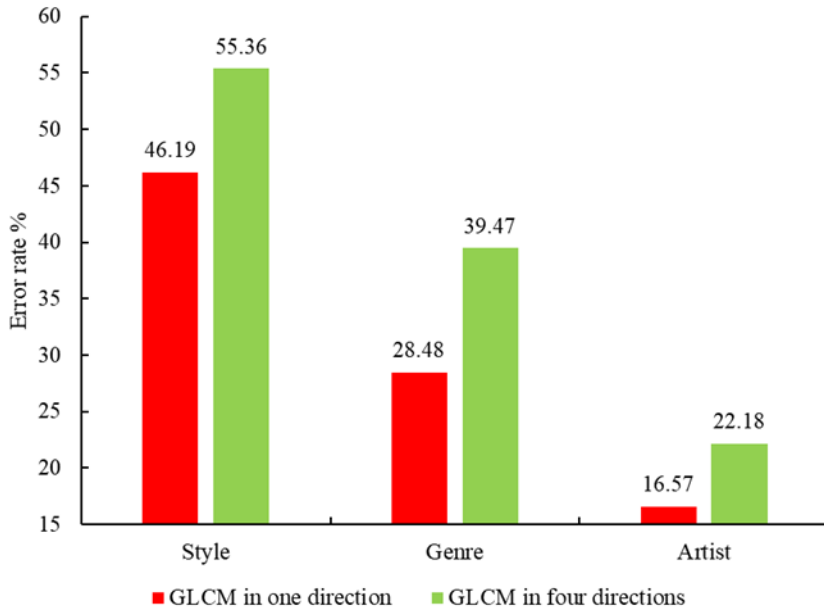
Figure 9. Comparison of Classification Performance With 50-Layer ResNet Stroke Representation



such as feature pyramids or convolutional neural networks can be used. The detailed experimental results are shown in Figure 9 and Figure 10.

Unlike the experimental results on the stroke information channel, the classification performance of the GLCM in four orientations is inferior to that of the single orientation in the two-channel model. The dual-channel model utilizes both stroke information and color information for classification. The advantage of this method is that it can comprehensively utilize the two main features of the image, thus providing a more comprehensive description of the artwork. The dual-channel model can better capture the details and overall style of painting works, thereby improving the accuracy

Figure 10. Comparison of Classification Performance With 131-Layer DPN Stroke Representation Method

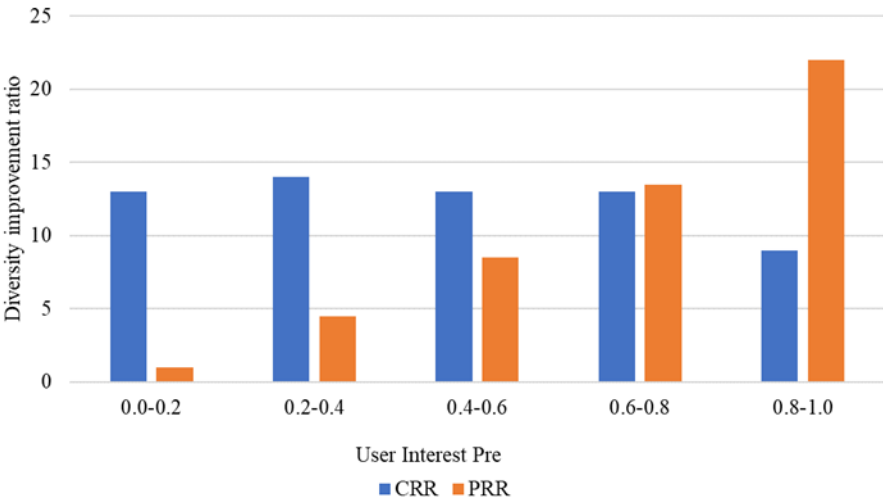


of classification. In addition to the stroke channel and dual-channel models mentioned above, there are many other stroke representation methods, such as stroke direction histograms, stroke length histograms, and so on. These methods each have their own emphasis on extracting stroke features and may exhibit different performance in different classification tasks. The experimental results in the figure are inconsistent with the previous experimental results. Although the four-direction GLCM achieves better classification performance in the stroke channel, it does not perform as well as the single-direction GLCM in the two-channel model.

In the process of recommendation system development, various recommendation algorithms have been proposed by researchers, and they have shown good recommendation effects in suitable application scenarios. Most of the recommendation algorithms are dedicated to the high accuracy of the recommendation results and strive to meet the user's preferences to the greatest extent possible. However, in practical recommendation applications, the purpose of personalized recommendation for users is to greatly enhance user satisfaction. Although the current research has not been able to provide a general and universal evaluation standard for user satisfaction, high accuracy is not the only criterion for enhancing user satisfaction. Performance is also an important indicator for evaluating a recommendation algorithm. A good recommender system should have good accuracy and variety at the same time. The diversity-oriented recommendation list personalized re-ranking algorithm (PBR) proposed in this paper is compared with the traditional re-ranking algorithm (CRR), and the results are shown in Figure 11.

It is obviously found that the accuracy indicators of the PRR algorithm have less loss than those before the reordering, and the accuracy loss of the former is acceptable compared to the larger reduction of the CRR algorithm. In the experiment, we start from the traditional diversity indicators and compare the results of the CRR and PRR algorithms. It can be found in the figure that the CRR has the same increase in the diversity of user recommendation results for each interest degree interval, and PRR is based on the degree of interest to improve the diversity of different degrees; the low degree of interest in the user diversity increase is small, while the high degree of interest users greatly increase the diversity.

Figure 11. Percentage Improvement of the Diversity of Recommendation Results for Users With Different Degrees of Interest



Analysis of Practical Applications

When exploring the innovative model of combining mobile learning with folk art education, we must recognize that this concept also has some limitations. Just like any educational model, mobile learning also faces challenges in technology, resources, personalized learning, and teaching quality supervision in folk art education. A deeper understanding of these limitations helps us to have a more comprehensive understanding of the current application status and development prospects of this new model. Although this article explores a new model of combining mobile learning with folk art education, there are also some limitations:

- Technical limitations: Mobile learning relies on mobile Internet technology and intelligent devices. In some regions or groups, due to insufficient network coverage or equipment conditions, it may not be possible to fully realize the application of mobile learning, which will limit the popularity and implementation of this model. Provide technical training and support to help teachers and students fully utilize mobile learning platforms, including training on intelligent device operation and network technology application, to ensure the smooth implementation of mobile learning.
- Insufficient educational resources: Although mobile learning can enrich teaching content and resources, the application effect of mobile learning in folk art education may be limited without high-quality educational resources and professional teaching team support. Establish a platform for sharing folk art education resources and integrate and share high-quality educational resources, including video tutorials, learning materials, art works, and so on, to enrich learning content and improve teaching effectiveness.
- Difficulty in achieving personalized learning: Although mobile learning can achieve personalized teaching, due to the particularity of folk art education, personalized learning may face challenges because it needs to consider factors such as learners' aesthetic background and cultural habits, which may increase the difficulty of personalized learning. Through intelligent technology and personalized learning systems, personalized learning content and paths are provided based on the learning style, interests, and progress of learners to meet the needs of different learners.
- Difficulty in regulating teaching quality: In the mobile learning mode, the supervision and evaluation of teaching quality may face certain difficulties. How to ensure that learners

receive high-quality folk art education is still a problem that needs to be solved. Establish a sound mechanism for evaluating and supervising teaching quality, including the development of evaluation indicators, the design of evaluation tools, and the supervision of the teaching process, to ensure the effective implementation of mobile learning in folk art education and the improvement of teaching quality.

By taking these measures, we can effectively overcome the limitations of combining mobile learning with folk art education, promote the application and development of this new model, and inject new vitality and momentum into the reform and innovation of folk art education. The application of this article in real life mainly includes the following aspects.

- Construction of mobile learning platforms: Governments, educational institutions, or folk art organizations can establish mobile learning platforms to provide online courses, teaching resources, and learning communities for folk art education, allowing students to learn anytime and anywhere through mobile phones or tablets.
- Exploration of innovative teaching models: Teachers can use mobile learning platforms to explore innovative teaching methods and forms, such as combining video tutorials, interactive games, virtual reality experiences, and so on, to improve teaching effectiveness and attract student interest.
- Provision of personalized learning experience: Through mobile learning platforms, students can choose appropriate courses and learning paths based on their interests, hobbies, and learning needs; enjoy a personalized learning experience; and improve learning motivation and effectiveness.
- Sharing and exchange of educational resources: Educational institutions and teachers can share high-quality teaching resources and experiences on mobile learning platforms, promote the sharing and exchange of educational resources, and improve the level and quality of the entire folk art education.

Through the above applications, the combination of mobile learning and folk art education can be effectively promoted, improving the popularity, quality, and effectiveness of folk art education and promoting the inheritance and innovative development of folk art traditions. The following development directions can be considered in the future.

- Application of intelligent technology: Further integrate artificial intelligence, virtual reality, augmented reality, and other technologies to provide a richer and more intelligent learning experience for folk art education, such as personalized learning recommendations and customized teaching plans through intelligent systems.
- Cross platform integration: Combine mobile learning with traditional educational forms, explore the integration mode of online and offline teaching, and provide students with more diverse learning choices and experiences.
- International cooperation and exchange: Strengthen international cooperation and exchange, draw on advanced international mobile learning and art education experience, and promote the international development of folk art education.
- Teaching content innovation: Continuously innovate teaching content and methods, combine contemporary art development trends and student needs, design teaching content that meets the characteristics of the times and personalized needs.
- Data driven teaching optimization: Utilizing big data and data analysis techniques to track and analyze student learning behavior and effectiveness, provide data support and decision-making reference for teaching optimization.

By continuously exploring and innovating, combining mobile learning with folk art education can better adapt to the needs of the times, improve the quality and level of folk art education, and promote the sustainable development and progress of art education.

CONCLUSION

This article aims to explore a new model that combines mobile learning with folk art education and apply it to educational practice. By studying and summarizing the application of mobile learning in folk art education, the following conclusions can be drawn. Firstly, mobile learning brings new possibilities and opportunities for folk art education. Traditional folk art education is limited by time, location, and resources, and mobile learning can break these limitations, allowing learners to access the necessary learning content and resources anytime and anywhere, improving the convenience and flexibility of learning. Secondly, mobile learning has promoted the innovation and reform of folk art education models. The traditional teaching model often focuses on imparting knowledge, lacking interactive and practical elements. However, mobile learning can enrich teaching content, enhance learner engagement and learning effectiveness through multimedia technology, interactive learning, and personalized teaching. Once again, mobile learning cultivates learners' comprehensive qualities and innovative abilities. Through mobile learning, learners can not only acquire knowledge and skills related to folk art but also cultivate comprehensive qualities, such as hands-on ability, aesthetic ability, and innovation ability, improving their competitiveness and application ability in the field of folk art. Finally, mobile learning provides new avenues and support for the inheritance and development of folk art. With the continuous development and popularization of mobile Internet technology, mobile learning has become a new way of learning, which can promote the inheritance and innovation of folk art and inject new vitality and power into the development of folk art. In summary, the combination of mobile learning and folk art education has important significance and value, which helps to promote the inheritance and development of folk art, promote people's understanding and appreciation of folk art, and promote the development and dissemination of cultural diversity. Therefore, we should actively advocate and promote a new model that combines mobile learning with folk art education and make greater contributions to the prosperity and development of folk art.

DATA AVAILABILITY

The figures used to support the findings of this study are included in the article.

CONFLICTS OF INTEREST

The authors declare that they have no conflicts of interest.

FUNDING STATEMENT

This work was supported by the Research and Practice Project on Higher Education Teaching Reform in Henan Province in 2024: Research and Practice on the Cultivation Path of “New Liberal Arts” Talents Based on Intangible Heritage Inheritance and Innovation

PROCESS DATES

Received: This manuscript was initially received for consideration for the journal on 02/18/2024, revisions were received for the manuscript following the double-anonymized peer review on 05/17/2024, the manuscript was formally accepted on 04/26/2024, and the manuscript was finalized for publication on 05/17/2024

CORRESPONDING AUTHOR

Correspondence should be addressed to Xiangqun Wang; wxqun928@126.com

REFERENCES

- Abayomi-Alli, O. O., Damaševičius, R., Misra, S., & Maskeliūnas, R. (2021). Cassava disease recognition from low-quality images using enhanced data augmentation model and deep learning. *Expert Systems: International Journal of Knowledge Engineering and Neural Networks*, 38(7), e12746. Advance online publication. 10.1111/exsy.12746
- Alkaf, M., Hartanto, H., Syahrial, S., & Rokhim, N. (2024). Folk art teaching as the element to create local-wisdom based character: A case studi on Buto Birowo kid dance group. *Asian Journal of Management, Entrepreneurship and Social Science*, 4(01), 1185–1196. <https://www.ajmesc.com/index.php/ajmesc/article/view/675>
- Gong, Y. Y. (2024). Application of multimedia data feature extraction technology in folk art creation. [IJIT]. *International Journal of Intelligent Information Technologies*, 20(1), 1–14. 10.4018/IJIT.340939
- Gryglewski, P., Ivashko, Y., Chernyshev, D., Chang, P., & Dmytrenko, A. (2020). Art as a message realized through various means of artistic expression. *Art Inquiry*, 22, 57–88. 10.26485/AI/2020/22/4
- Kalota, F. (2024). A primer on generative artificial intelligence. *Education Sciences*, 14(2), 172. Advance online publication. 10.3390/educsci14020172
- Kitto, S., Ng, Y. Y., & Cleland, J. (2024). ‘Slowing down when you should’: Optimising the translation of artificial intelligence into medical school curricula. *Singapore Medical Journal*, 65(3), 186–189. 10.4103/singaporemedj.SMJ-2023-19038527305
- Li, D., & Zhang, Y. (2020). Multi-instance learning algorithm based on LSTM for Chinese painting image classification. *IEEE Access : Practical Innovations, Open Solutions*, 8, 179336–179345. 10.1109/ACCESS.2020.3027982
- Li, F. (2024). Chord-based music generation using long short-term memory neural networks in the context of artificial intelligence. *The Journal of Supercomputing*, 80(5), 6068–6092. 10.1007/s11227-023-05704-3
- Liu, Y. (2020). Symbol analysis of intangible heritage Miao paintings in Xiangxi of China. *Research Square*. 10.21203/rs.3.rs-74149/v1
- Liu, Y. (2023). On the value and significance of folk art in art education in colleges and universities. *Educational Review*, 7(6), 730–733. 10.26855/er.2023.06.015
- North, C., Hills, D., Maher, P., Farkić, J., Zeilmann, V., Waite, S., Takano, T., Prince, H., Gurholt, K. P., Muthomi, N., Njenga, D., Karaka-Clarke, T. H., Mackenzie, S. H., & French, G. (2024). The impact of artificial intelligence on adventure education and outdoor learning: International perspectives. *Journal of Adventure Education and Outdoor Learning*, 24(1), 123–140. 10.1080/14729679.2023.2248302
- Ochilova, N. R. (2024). Means of moral education of students. *Oriental Renaissance: Innovative, Educational, Natural and Social Sciences*, 4(3), 428–438. https://oriens.uz/media/journalarticles/56_Ochilova_Nigora_Ruzimuratovna_428-438.pdf
- Rahmatjonzoda, A. (2024). The need to improve the ethno-pedagogical system of preparing future pedagogues for spiritual education activities. *Theoretical Aspects in the Formation of Pedagogical Sciences*, 3(3), 106–112.
- Rochovská, I., & Švábová, B. (2021). The use of the interpretation of works of art in pre-school education. *Problems of Education in the 21st Century*, 79(6), 956–970. 10.33225/pec/21.79.956
- Sade, A., Tiwari, V., & Kesaboyina, V. M. (2024). The role of leather puppetry art in preschool: An exploratory study on government schools in Andhra Pradesh. *Educational Administration: Theory and Practice*, 30(4), 572–580. 10.53555/kuey.v30i4.1506
- Shodieva, M. (2024). The influence of artificial intelligence on education. *Modern Science and Research*, 3(2), 394–401. 10.5281/zenodo.1064759
- Wang, Z. (2024). Art and design education and factors of technological revolution. *Journal of Education and Educational Research*, 7(3), 179–185. 10.54097/Osr5b921
- Wang, Z., Cui, R., Cong, T., & Liang, H. E. (2021). Overseas dissemination of ancient Chinese costume culture from the perspective of cultural confidence. *Fibres & Textiles in Eastern Europe*, 3(147), 111–116. 10.5604/01.3001.0014.7796

Wei, Y. (2021). Digital inheritance application of Chinese traditional folk art in art teaching in colleges and universities. *Forest Chemicals Review*, 103–109. 10.17762/jfcr.vi.57

Wu, S. N., & Sensai, P. (2021). *Application of Tujia folk songs in music teaching of Tongren preschool education college* (Master's thesis, Mahasarakham University). Mahasarakham University Intellectual Repository. <http://202.28.34.124/dspace/handle123456789/897>

Zhang, M., & de Bont, C. (2023). Exploring the sustainability of Huizhou bamboo carving in the perspective of heritage craft. *FormAkademisk*, 16(4). Advance online publication. 10.7577/formakademisk.5451

Zhong, S. H., Huang, X., & Xiao, Z. (2020). Fine-art painting classification via two-channel dual path networks. *International Journal of Machine Learning and Cybernetics*, 11(1), 137–152. 10.1007/s13042-019-00963-0

Zhu, J. (2023). Research on the fusion strategy of traditional arts and crafts communication and short video. *The Frontiers of Society. Science and Technology*, 5(3), 121–126. 10.25236/FSST.2023.050319

Xiangqun Wang was born in Henan, China, in 1981. From 2009 to 2010, she studied in Belarusian State University of Culture and Arts, and received her Master's degree in 2010. She is a representative inheritor of the intangible cultural heritage of Ru porcelain firing techniques, a member of the Youth Committee of the Chinese Folk Literature and Art Association, and a first-class technician in ceramic decoration. Currently, she works in Henan Finance University, and her research interests are included art and design, ceramic art.